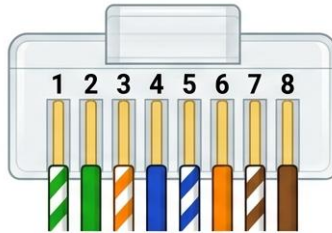


Ethernet Standard and Specification

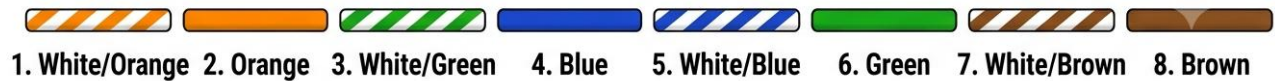
Task 1: RJ-45 Pin Layout and T568A/T568B Standards



T568A COLOR CODE



T568B COLOR CODE



Task 2: Comparison of Ethernet Standards

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3 or better twisted-pair	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better twisted-pair	100 m	RJ-45
1000BASE-T	1 Gbps	Cat5e or better twisted-pair	100 m	RJ-45

Task 3: Connecting Two Laptops Directly

For a traditional direct Ethernet connection between two laptops, I would use a crossover cable. A crossover cable swaps the transmit and receive wire pairs, so the transmit pins of one laptop connect to the receive pins of the other laptop. However, most modern laptops support Auto-MDI/MDI-X, so they can often work with a normal straight-through cable as well.

Task 4: Ethernet Devices at Home or College

Device	Where it is used	Likely Ethernet Standard
Wi-Fi Router	Home	1000BASE-T
Desktop Computer	Home/College	1000BASE-T
Network Switch	College computer lab	1000BASE-T